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FEDERAL SPECIFICATION

DISTILLATION UNITS, WATER, THERMOCOMPRESSION, DIESEL-ENGINE-DRIVEN, SKID- OR TRAILER-MOUNTED (60 to 1000 GPH)

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for diesel-engine-driven water distillation units.

1.2 Classification. Distillation units shall be of the following classes as specified (see 6.2).

Class 1 - Skid-mounted; with capacities of 60 to 1000 gallons per hour (gph) (227 to 3785 litre per hour (Lph)).

Class 2 - Trailer-mounted; with capacities of 60 to 125 gph (227 to 473 Lph).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 1581), Naval Construction Battalion Center, Port Hueneme, CA 93043-4301, by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4620

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Federal Specifications

L-P-391	- Plastic Sheets, Rods and Tubing, Rigid Cast, Methacrylate (Multiapplication).
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths.

Federal Standard

FED-STD-595	- Colors Used in Government Procurement.

Commercial Item Description

A-A-55057	- Panels, Wood/Wood Based; Construction and Decorative.

Military Specification

MIL-C-45150	- Chassis, Trailer, 2-Wheel, Cart Type 1/4 to 3-1/2 Ton.

Military Standard

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions are obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.)

2.1.1 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

C63.5 - Standard for Electromagnetic Compatibility - Radiated Emission Measurements in Electromagnetic Interference (EMI) Control - Calibration of Antennas.

(Private sector and civil agencies may purchase copies of these voluntary standards from the, American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.)

ASTM

ASTM B 111 - Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock.
ASTM B 209 - Aluminum and Aluminum-Alloy Sheet and Plate.
ASTM B 543 - Welded Copper and Copper-Alloy Heat Exchanger Tube.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

MANUFACTURER'S STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-25 - Standard Marking System for Valves, Fittings, Flanges, and Unions.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Manufacturer's Standardization Society of the Valve and Fittings Industry, 420 Lexington Avenue, New York, NY 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.
SAE J534 - Lubrication Fittings, Standard.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 11006, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The distillation unit shall be complete in all details and shall be constructed as specified herein. The design and materials shall be such that the entire unit is of minimum weight consistent with the ruggedness, serviceability, and accessibility necessary for military use. Strength and quality shall not be sacrificed for lightness in weight. The distillation unit shall consist essentially of one or more of the following component parts: an engine; evaporator; vapor compressor; blowdown (overflow) heat exchanger; distillate heat exchanger; exhaust gas heat exchanger; deaerator for oxygen removal from feedwater; blowdown and distillate pumps; piping; valves; controls; flow indicators and gages; as well as any other auxiliary equipment necessary to, and compatible with, the distillation unit design and requirements for satisfactory operation. The distillation unit, as delivered, shall be complete and ready for operational use.

3.2 First article. When specified (see 6.2), the contractor shall furnish a distillation unit of the class and capacity as required for first article inspection and approval (see 4.2.1 and 6.5).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.4 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided wherever possible in order to minimize galvanic corrosion between these metals. When the use of dissimilar metals cannot be avoided, an interposing material which will minimize or eliminate this galvanic effect shall be used.

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3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Acid/scale inhibiting chemical injection system. Unless otherwise specified (see 6.2), the injection system shall be furnished and installed. Such a system shall include, but is not limited to:

a. A suitable nonfragile acid-proof container having a capacity not less than 1 gallon (3.78 litre).

b. Manual feeding of the container.

c. Means for positive injection of acid or scale inhibitor based on the quantity of raw water feed.

d. Design that precludes contamination of the distilled water.

e. An acid or chemical transportation system having pipe, valves, and fittings designed for acid injection without leakage.

f. Means ensuring safety in the handling and feeding of the acid.

g. Acid/chemical addition sufficient to induce the $HC0_3$, to retard $CaC0_3$, scale formation and yield a pH between 7.2 and 7.5 after deaeration.

3.6.1 Automatic recycling. When specified (see 6.2), the acid-injection system shall be designed to permit the acid to be automatically recycled and heated until the evaporator tubes are clean.

3.7 Operating characteristics.

3.7.1 Output. The distillation unit shall be capable of operating continuously on normal natural seawater except for short shutdown periods. During such normal operation the average rate of distillation shall be not less than the capacity specified. The clean tube economy in pounds (lb) (0.453 kilogram (kg)) of distillate produced per lb (kg) of fuel used shall not be less than 300 to 1. In no event shall economy be less than 250 to 1. At any time economy is less than 300 to 1, the unit shall be capable of being readily cleaned, without the disassembly of any part, and having the economy return of 300 to 1 or greater.

3.7.2 Chemical contents. The distillation units shall produce drinking water with a sodium chloride content not over 17.2 parts per million (ppm), a copper content not over 1 ppm, and zinc content not over 3 ppm.

3.7.3 Methods of scale removal. The design shall be such as to facilitate chemical retardation and removal and, when practicable, mechanical removal of scale from all heat exchanger surfaces.

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3.7.4 Blowdown. When operation involves normal seawater, the blowdown shall be designed to operate with 1.8:1 or 1.5:1 concentration ratio. However, the blowdown rate shall at all times be sufficient to maintain a brine concentration of less than 3/32.

3.7.5 Water temperature. The distillation unit shall meet the operating requirements at a raw feedwater temperature of not less than 30 degrees Fahrenheit (°F) (-1 degrees Celsius (°C)), and when using raw feedwater at a temperature of 70 °F (21 °C), shall produce distillate with a temperature not in excess of 85 °F (29 °C). The equipment furnished shall include raw feedwater and distillate temperature gages of an adequate number, and installed in appropriate locations, to permit continued observation of operating temperatures to assist operating personnel in their evaluations of those equipment operational efficiencies used as a measure of the need for routine maintenance actions such as internal cleaning.

3.8 Capacity. The distillation unit shall have the blowdown rate specified in 3.7.4 and shall produce distillate as specified in 3.7.1.

3.9 Power plant.

3.9.1 Engine. The manufacturer's standard commercial diesel engine shall be liquid-cooled, compression ignition, and shall be furnished with the following: keyless electric cranking system, engine shutoff type lubricating oil control device, charging alternator with capacity rating compatible to the system, and instrumentation. The instrumentation shall be those gages, meters, and indicators normally supplied with the manufacturer's standard product, but as a minimum, shall include a lubricating oil pressure gage, tachometer-hourmeter, and battery charging indicator. Starting aids shall be either electric glow plug or ether primer. When ether primer is required, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces (355 millilitre).

3.9.1.1 Power speed rating. The power and speed rating shall be such that operation of the distillation unit at rated capacity shall not require horsepower (hp) (watt (W)) in excess of the continuous brake hp (W) rating of the engine for the applicable speed as established by the engine manufacturer.

3.9.1.2 Electrical cranking system. The system shall achieve self-sustained combustion at the manufacturer's rated idle speed in less than 5 minutes in ambient temperatures from 20 °F to 125 °F (-6.6 °C to 51.6 °C).

3.9.1.2.1 Storage battery(s). The manufacturer's standard commercial storage battery(s) in the unit shall be dry charged, without electrolyte, and shall be furnished with sealed caps to prevent atmospheric moisture from entering the battery.

3.9.1.3 Fuel system. The fuel system shall provide for continuous operation with fuel supplied directly from the containers in which the fuel is delivered. The system shall consist of two fuel supply lines, a return line, and a three-way brass valve so arranged that fuel may be taken from

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either of two containers, and both lines may be shut off simultaneously. The fuel lines shall be at least 6 feet (1828 millimetre (mm)) in length. The three-way valve and fittings shall have 0.25-inch National taper pipe threads (NPT).

3.9.1.4 Cooling system. The engine shall be equipped with a pressure cooling system that uses the distillate as a coolant.

3.10 Heat transfer components.

3.10.1 Evaporator. The evaporator shall be designed of tubular unit or plate heat exchanger for easy removal and replacement and shall be designed for natural circulation of the sea water so that the volume of free water shall be at a minimum. Unless otherwise specified (see 6.2), tube material shall be 90/10 or 70/30 copper-nickel alloy not less than 0.035-inch (0.899 mm) nominal wall thickness in accordance with ASTM B 111 for seamless tubes and ASTM B 543 for welded tubes. Plate heat exchanger shall be titanium. The vapor disengagement space shall be provided with two protected sight glasses of sufficient viewing area, as well as an evaporator liquid level gage, located to permit observation of the water level and boiling action. The evaporator shall be provided with lifting and handling attachments, and shall be so designed that all tubes are accessible for descaling, removal, replacement, or plugging.

3.10.2 Blowdown and distillate heat exchanger. The heat exchanger shall be of a stream plate type, so constructed as to allow removal or replacement by sections or components. Unless otherwise specified (see 6.2), heat exchanger plate material shall be titanium. Drain cocks shall be provided to sample the distillate. Inlet to the blowdown heat exchanger shall be provided with a strainer 40 to 60 mesh and so arranged that strainer can be flushed clean to atmosphere while unit is in operation.

3.10.3 Exhaust gas heat exchanger. The exhaust gas heat exchanger shall utilize the engine cooling water and shall be so designed that the exhaust gases pass through the tubes. Material shall be corrosion- and erosion-resistant. The covers shall swing open not less than a full 90 degrees to make the tubes accessible for cleaning or replacement.

3.11 Compressor. The compressor shall be of the rotary type, or centrifugal, heavy-duty type designed for continuous operation at speeds recommended by the manufacturer. All parts of the compressor in contact with the steam shall be of corrosion-resistant steel, aluminum, or other compatible corrosion- and erosion-resistant materials. Steam shaft seals shall be mechanical and employ a carbon seal face. Lubrication shall be by a forced flow system suitable for high temperature operation and shall preclude contamination of the steam by the lubricant. An oil level sight glass shall be provided on the oil reservoir.

3.12 Pumps. Unless otherwise specified (see 6.2), pumps shall be of the centrifugal type. Pump volutes and rotors shall be of nickel aluminum bronze construction, with not less than 3 percent nickel, have shafts of monel, and be equipped with shaft seals to prevent leakage around the shaft. Seal casing and springs shall be monel. Pump connection shall be flanged or internally threaded

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NPT. Pumps shall be belt-, shaft-, or gear-driven from the engine. The distillation unit shall be provided with feedwater, blowdown, and distillate pumps.

3.13 Power transmission. The means of power transmission employed in the unit shall comply with the following general characteristics.

3.13.1 Belting. Belting shall be of the V-belt, endless type, shall have an operating life of not less than 2000 hours under field conditions, shall be replaceable without dismantling any integral part of the unit, and shall be provided with means for tightening. Tightening shall be accomplished by use of a flat idler pullup on the outside, a V pullup on the inside, or on movable mounts.

3.13.2 Other drive assemblies. All parts of the drive assemblies, except belting and bearings, shall be designed for an operating life of not less than 15 000 hours.

3.14 Strainer. A duplex, brass, feed-water strainer having two 60-80 mesh baskets shall be permanently mounted on the unit. The assembly shall have a valve arrangement to permit one basket to be used while the other is cleaned. Strainer caps shall be provided with yoke-type clamps.

3.15 Hose, pipe, fittings, and connections. Thickwall reinforced hose may be used for connecting lines in which temperatures are not expected to exceed 230 °F (110 °C), however, the use of hose should be kept to a minimum. Suitable stainless steel clamps shall be used at hose connections. Piping, where used, shall be nonferrous corrosion-resistant with silver alloy sweat type joints, except that engine exhaust pipe shall be steel and fittings in distilled water lines may be installed with 50/50 solder. The number of sizes of pipe and hose shall be kept to a minimum. Valves and fittings shall be bronze. Manual flow control valve shall be globe or angle type. Manual shutoff valves shall be gate type. Hose fittings shall be marked in accordance with the manufacturer's standard practice or with MSS-SP-25.

3.15.1 Drains. Complete drainage of all parts or components shall be provided by means of suitably located gate valves, cocks, or plugs. Drainage time of any part or components shall be not more than 90 minutes. Unless otherwise specified (see 6.2), all drain points shall be painted bright yellow, matching color 13538 in accordance with FED-STD-595.

3.16 Controls and control instruments.

3.16.1 Electromagnetic interference suppression. When specified (see 6.2), the distillation unit shall conform to the electromagnetic interference suppression requirements of ANSI C63.5.

3.16.2 Controls. All controls, both manual and automatic, necessary or advisable for normal or emergency operation shall be installed in all lines or systems. Means for control shall be provided for positive regulation of the blowdown to insure an adequate flow at all times. Raw water feed shall be controlled to maintain a constant level in the evaporator. A diaphragm valve actuated by changes in steam pressure shall maintain boiling pressure at 1 pound-force per square inch gage

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(psig) ±0.5 psig (0.070 kilograms-force per square centimetre gage (kgf/cm^2) ±0.035 kgf/cm^2) in the evaporator bypassing cold feed water to the evaporator.

3.16.3 Control instruments. When practicable, instruments shall be mounted on a vibration dampened control panel in an accessible location easily readable by the operator, except that direct reading thermometers shall be located to obtain maximum accuracy and shall be easily readable. Thermometers shall be marked in °F (°C) and calibrated at 212 °F (100 °C). Gages shall be constructed of corrosion-resistant materials and shall be vibration proof.

3.17 Housing. All units shall be furnished with a suitable housing for protection from wind and weather. The top and supporting legs shall be of aluminum conforming to ASTM B 209, temper T-4. Legs shall be easily removable and shall be held by bolts or pins. Side cover panels shall be made of metal-reinforced sheet plywood or of canvas. Plywood shall conform to NN-P-530, group B. exterior type, grade A-C, not less than 0.50-inch (12.7 mm) thick. The panels shall be equipped with windows of clear plastic conforming to L-P-391, grade 1, not less than 0.10-inch (2.5 mm) thick. The windows shall be situated for clear view of all instruments, gages and sight glasses. The panels shall be easily demountable for access to the unit. The panels shall be hinged at the top and approximately at midpoint to permit their being folded up to form a canopy. The canopy, when in the up position, shall have a vertical clear height from the underside of the mounting of not less than 6.75 feet (2 057 mm). If canvas fabric is used, it shall be mildew-resistant and waterproof treated in accordance with CCC-D-950. Devices for securing the panels in the down and in the rolled up positions shall be provided.

3.18 Thermal insulation. Components or parts of the distillation unit shall be suitably insulated. Insulation of component parts, when necessary, shall be the fixed type consisting of glass fibers, magnesia block with canvas covering, suitably treated, or other appropriate material. Insulation will be so designed as not to interfere with disassembly or assembly of parts or shall be the removable type of glass fiber or other suitable thermal-insulating rigid material. Any lagging or metal support will be of aluminum conforming to ASTM B 209, temper T-4, 26 gage. Canvas covering for insulation shall be waterproofed and fungus-proofed in conformance to CCC-D-950.

3.19 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1000 pound-force per square inch (70.3 kgf/cm^2) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.20 Mounting for class 1 (skid-mounted) units. The skid shall be of structural steel shapes gusseted and braced to prevent sagging or distortion of the unit. Ends of the skid shall be turned up to facilitate skidding and provided with pulling eyes or rings having an opening of not less than 3 inches (76.2 mm). The eyes of the skid shall withstand operational pulling forces without failure or deformation and may be reinforced transversely with pipe having an inside diameter not less than 3 inches (76.2 mm). The engine and compressor shall each be mounted on the skid frame.

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3.21 Mounting for class 2 (trailer-mounted) units. Unless otherwise specified (see 6.2), the unit shall be mounted on a standard military 3.5-ton (3 556 kg) trailer, chassis-model 353. The trailer chassis shall be manufactured in accordance with MIL-C-45150. The contractor may alter the trailer frame for the most suitable mounting of the equipment and the lowest possible center of gravity. The trailer shall be equipped with four stabilizers on the corners of the trailer, two at the rear and two at the front. A spring loaded, military-type pintle hook shall be mounted on the rear center of the vehicle.

3.22 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 0.0025-inch (0.065 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.23 Stenciling. The gross weight of each unit of equipment shall be stenciled on each side of the unit in a manner as to be readily discernible to dock and military personnel. The prescribed tire pressure shall be stenciled on the frame or body near the wheels with block- or stencil-type letters not more than l-inch (25.4 mm) high.

3.24 Technical publications. When specified (see 6.2), technical publications shall be furnished and shall include at least a manual containing complete installation, operating, and maintenance instructions. The manual shall also contain complete parts lists that indicate the name, model number, parts number, number of parts per unit, and parts costs prevailing at time of contract award of both the manufacturer's suppliers and the manufacturer.

3.25 Instruction plates. The distillation units shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 0.125-inch (3.175 mm) diameter.

3.26 Identification marking. Identification shall be permanently and legibly marked directly on the distillation units or on a corrosion-resisting metal plate securely attached to the distillation units at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.27 Lifting and tiedown attachments. When specified (see 6.2), the distillation units shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to

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type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the distillation units. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the distillation units on the carrier when shipped.

3.28 Repair parts and maintenance tools. When specified (see 6.2), repair parts and maintenance tools shall be furnished.

3.28.1 Toolbox. When specified (see 6.2), the equipment shall be provided with a toolbox made of metal not less than 0.0747-inch (1.897 mm) in nominal thickness, or of wood construction. The toolbox shall have a hinged lid and truck drawbolt of a type that will keep the lid closed during vibration or rough travel. The toolbox shall be securely fastened to the equipment.

3.29 Items to be furnished with prime equipment. No shipment of prime equipment shall be made unless repair parts and maintenance tools as specified (see 3.28) are included with the shipment, or unless written approval for shipment of the prime equipment without such items has been received from the systems command or service concerned.

3.30 Workmanship.

3.30.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.30.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.30.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.30.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.30.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.31 Operating safety. All rotating and reciprocating parts, or parts subject to high temperature, having inherent qualities or location that makes them hazardous to operating personnel shall be insulated, fully enclosed, or properly guarded. Such guards may be fabricated of solid or expanded sheet steel and shall be firmly attached to the framework by means of bolts or screws. In addition, the equipment furnished shall incorporate overall safety features to satisfy all applicable OSHA Standard 1910.219 provisions relative to personnel safety during operation.

3.32 Occupational noise exposure. The equipment furnished shall not subject operating personnel to a noise level in excess of 100 decibels (dB) absolute over an operating period of up to 2 hours, nor a noise level of over 90 dB absolute when operating continuously for up to 8 hours, and shall comply with OSHA 1926.52 provisions relative to occupational noise exposure criteria.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one complete unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the tests of 4.4.

4.3 Examination. Each distillation unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Each distillation unit shall be tested as specified in 4.4.1 and 4.4.2. The first article unit shall be tested as specified in 4.4.1 through 4.4.4. Failure to pass any test shall be cause for rejection of a unit.

4.4.1 Tests during assemble. All heat transfer components shall be pressure tested for leaks before being assembled into the distillation unit.

4.4.2 Capacity test. Each completed distillation unit shall be operated at the place of manufacture for a 4-hour period of continuous, level operation at rated capacity as in 3.7.1 with use of 3.5 percent sodium chloride solution at a temperature of 50 °F to 75 °F (10 °C to 23 °C) Distillate produced and fuel consumed shall be measured to determine normal clean tube operating efficiency (see 3.7.1 and 3.7.2). Copper/zinc content shall be measured only in first article.

4.4.3 Lifting and tying down attachment tests. When required (see 6.2), the lifting and tying down attachments shall be tested to conform to 3.27.

4.4.4 Performance tests. Upon completion of the foregoing tests, the first article unit shall be subjected to an operational performance test as listed below. The operating characteristics of the unit will be determined and recorded to determine conformance to this specification. Any modification to achieve the required performance shall be at the expense of the contractor.

a. The unit shall be operated, using seawater of not less than 35 000 ppm total dissolved solids, for a total period of 400 hours in sequence as follows:

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(1) With acid/chemical: One 2-hour run with raw feed water temperature at 40 °F (4 °C). One 8-hour run with raw feedwater temperature at 125 °F (51 °F). Three 8-hour runs with raw feedwater temperature at 70 °F (21 °C). 116 hours without temperature control of feed water to complete a total of 150 hours of operation. Operation may be continuous. Verification of capacity and economy shall be in conformance to 3.7.1.

(2) Without acid/chemical injection: 200 hours of continuous operation during which time maintenance of the unit shall be limited to lubrication, servicing, and adjustment of the various components. The total quantity of distillate produced and fuel consumed shall be measured to determine conformance to 3.7.1 for normal and minimum rated economy.

(3) With acid/chemical injection: 50 hours as part of the cleaning operation in preparation for the capacity test specified in 4.4.2.

b. A chemical test of the distillate shall be made to determine conformance to the requirements of 3.7.2.

c. Blowdown shall be measured to determine conformance to 3.7.4.

d. Final distillate temperature, when using raw feed water at a temperature of 70 °F (21 °C), shall, be checked to determine conformance to 3.7.5.

e. The condition of the insulation at the conclusion of all tests shall be noted to ensure that no deterioration or damage has occurred (see 3.18).

f. Upon satisfactory completion of the above tests, the distillation unit shall be drained, cleaned, and reoperated in accordance with 4.4.2.

4.4.5 Measurement of electromagnetic radiation. To determine conformance to the electromagnetic interference characteristics of 3.16.1, electromagnetic radiation shall be measured in accordance with ANSI C63.5. When suppressed to conform to the electromagnetic interference requirements of 3.16.1, the manufacturer may upon approval of the contracting officer, furnish a certification in lieu of the test, that the distillation units meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES

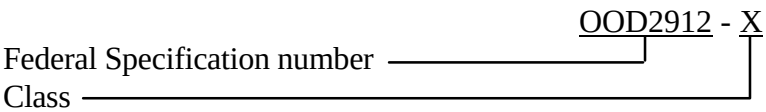
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The distillation units covered by this specification are intended for use in the conversion of seawater into potable drinking water.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class and capacity required (see 1.2).
c. When first article inspection is required and when a preproduction pack inspection is required (see 3.2 and 4.2.1).
d. When an acid/chemical injection system is not required (see 3.6).
e. When the acid/chemical injection system is to be designed to incorporate an automatic recycling feature (see 3.6.1).
f. Evaporator tube material if different (see 3.10.1).
g. Heat exchanger material if different (see 3.10.2).
h. When other than centrifugal type pumps are required (see 3.12).
i. When drain points are to be painted a color other than yellow (see 3.15.1).
j. When electromagnetic interference suppression is required (see 3.16.1).
k. When lubrication shall be other than the manufacturer's standard practice (see 3.19).
l. When the distillation unit shall not be mounted on trailer conforming to MIL-C-45150 (see 3.21).
m. Color of the finish coat required (see 3.22).
n. When technical publications are to be furnished (see 3.24).
o. When lifting and tiedown attachments are required (see 3.27).
p. When repair parts and maintenance tools are required (see 3.28).
q. When toolbox is required (see 3.28.1).

6.3 Part or Identifying Numbers (PINs). The specification number, class, type, and size are combined to form PINs for diesel-engine-driven water distillation units covered by this document (see 1.2). PINs for the diesel-engine-driven water distillation units are established as follows:



6.3.1 Class code number. The class of the distillation unit (see 1.2) is identified by a single digit number as follows:

Class 1: - 1 Class 2: - 2

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.6 Supersession data. This specification replaces Military Specification MIL-D-15691F dated 1 October 1991.

6.7 Classification cross reference. Classification used in this specification (see 1.2) are identical to those found in the superseded Military Specification, MIL-D-15691F.

6.8 Subject term (key word) listing.

Potable drinking water
Seawater

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue.

MILITARY INTERESTS:

Custodians:
Navy - YD1
Air Force - 99

Review Activities:
Navy - MC
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

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